
IK 506: Research methods and statistics for contemplative science

Course Code	IK 506
Course Name	Research methods and statistics for contemplative science
L-T-P-C	2-1-0-3
Prerequisites	None
Intended for	3rd & 4th Year B. Tech, Masters and PhD
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023); 42nd Senate (Jan 2024)

Course Contents

- Module 1 (21 Lectures + 7 tutorial Lectures) **Research methods theory:**
 - Basic assumptions underlying scientific research
 - Ethics in scientific research
 - Literature review and hypothesis formulation
 - Data collection methods
 - Measurement techniques & Sampling methods
 - Research designs
 - * Apart from controlled trial designs (including randomized controlled trial design-sRCT), emphasis will also be given on case-control study design and prospective cohort design from contemplative science perspective. For example, studying the effect of advanced meditation (with monks is more feasible from case-control design than RCT). Similarly naturalistic cohort long term follow-up studies are optimal from sampling perspective to study the effect of yogic/meditative lifestyle.
 - Procedure for conducting research experiment
 - Control techniques in experimental research
 - Mixed methods research
 - Emphasis on first person (for subjective experience) and third person perspective based assessments will be discussed. Special emphasis on experience sampling method and its relevance for contemplative science will be discussed
 - Scientific writing

Tutorial sessions

- Randomization procedure
- Scientific illustrations-Inkscape and blender
- Reference management-Zotero
- Qualitative data coding-Qualcoder
- Note: All the tutorial sessions will be taught with Yog/Meditation based dataset for better understanding of the concepts

Module 2 (7 Lectures+ 7 tutorial Lectures) Statistics theory

- Data representation-tables & figures
- Descriptive statistics
- Key ingredients for inferential statistics
- Hypothesis testing, statistical significance and decision errors
- T tests
- ANOVA
- Correlation Regression
- Chi square test
- Linear mixed model analysis (LMM)
- Distribution free statistics

Tutorial sessions (Using Jamovi & R-open-source free software)

- Data wrangling
- T tests
- ANOVA
- Correlation & Regression
- Chi square test
- LMM
- Sample size calculation-G power

Note: All the tutorial sessions will be taught with Yog/Meditation based dataset for better understanding of the concepts

Text Books

1. Christensen LB, Johnson B, Turner LA., **Research Methods, Design, and Analysis**, Pearson Education, 2019.
2. Aron A, Aron EN, **Statistics for psychology**, 6th Edition, Pearson Education, 2013 .

References

1. Zar JH, **Biostatistical analysis**, Pearson Education India, 1999.
2. Creswell JW, Poth CN, **Qualitative inquiry and research design: Choosing among five approaches**, Sage publications, 2016.